UNIVERSITI TEKNOLOGI MALAYSIA · FACULTY OF COMPUTING · SCSE2243 · APPLICATION DEVELOPMENT PROJECT I · SECTION 01 · SEMESTER II 2025/2026

UTM Borrow

A campus-wide circular resource-sharing platform — where every handover is cryptographically signed.

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N
NEED

Students buy once, use once, and have nowhere safe to pass it on

29 RESPONDENTS

UTM courses require specialised, single-use items — lab coats, scientific calculators, drawing sets, presentation attire — that lose all value after one semester. The only sharing channel that exists today is an unverified WhatsApp or Telegram group: **no identity, no reputation, and no record of who holds what**. Lenders are therefore afraid to lend, and the items sit unused in dorm rooms until they are thrown away.

62.1%
spend RM50–RM150 every year on items they use for a single semester

37.9%
say their biggest fear when lending is never getting the item back

31%
have no proof at all if a borrowed item comes back damaged

24.1%
never manage to find anyone willing to lend to them at all

Source: our own questionnaire, 29 UTM students across multiple faculties and years of study.

A
APPROACH

Every loan is a state machine; every handover is a signature

3 SUBSYSTEMS · 9 MODULES

Identity & User Management
Ahmat Mahamat Mourdji Moustapha

- Authentication
- Profile & Trust Score
- Activity Dashboard

Only `@utm.my` and `@graduate.utm.my` addresses may register. Passwords are bcrypt-hashed; sessions are signed JWTs. A 5-star Trust Score is recomputed from completed loans only.

Resource Catalog & Discovery
Muaz Ibne Ahmed

- Item Listing & Management
- Category & Condition Metadata
- Location-Based Filter

Every listing carries a server-enforced category and condition. Borrowers narrow by residential college and faculty, and sort by true great-circle distance over campus coordinates — no third-party map service.

Transaction & Handover
Mohammed Mohsen Mohammed Alsakkaf

- Request & Approval Workflow
- QR Verification
- Moderation & Reporting

A request atomically locks the item, so two students can never claim the same tool. Every state change is appended to an immutable log, and the admin portal is gated behind an authenticator code.

`Available` — Listed in the catalog
`Pending` — Request locks the item atomically
`Approved` — Lender accepts after reading the Trust Score
`Borrowed` — Lender scans the borrower's signed QR
`Completed` — Return scan closes it; both parties rate

PROOF OF PHYSICAL HANDOVER

`UTMB . <payload> . <signature>`

The **payload** binds the transaction, the borrower, a random nonce and the issue time. The **signature** is HMAC-SHA256 over it, computed with a server-held secret and re-verified on every scan. Tokens expire after **24 hours**, defeating replay of a captured code, and every attempt — success, expired, wrong transaction, already used — is written to a forensic scan log.

TECHNOLOGY, AS BUILT AND DEPLOYED

```
React 18 · Tailwind CSS · Framer Motion
FastAPI · Python 3.12 · uvicorn
MongoDB · Motor async driver
JWT HS256 · bcrypt · TOTP 2FA
HMAC-SHA256 QR · Server-Sent Events
Vercel + Render
```

27/27 use cases fully implemented — each with a working endpoint, its database collections, and the matching screen, all wired together and live.

B
BENEFIT

What the community gets

BORROWERS · LENDERS · THE UNIVERSITY

Borrowers stop buying

The **RM50–RM150** a student currently spends each year on single-use items stays in their pocket. Items are found by college and faculty, so collection is a walk, not a trip — and dorm rooms stay clear of things used once a year.

Lenders lend without fear

The counterpart is a verified UTM member with a Trust Score that had to be earned. If a dispute arises, there is **cryptographic proof of both the handover and the return**, timestamped — not a screenshot.

The university keeps control

The email gate keeps non-affiliates off campus trade entirely. Moderation is MFA-gated and every decision is permanently audited, so student welfare cases can be reviewed rather than argued.

Every item borrowed instead of bought is one fewer purchased, and one fewer discarded at the end of semester. The platform turns that into reportable data for UTM's **Smart & Green Campus** agenda and **UN SDG 12 — Responsible Consumption and Production**. In our survey, **69%** of students said an environmental-impact screen would make them want to share more.

C
COMPETITION

What makes it different from what students use today

UNIQUENESS

CAPABILITY	WHATSAPP & TELEGRAM GROUPS	MARKETPLACE APPS	FACULTY / LIBRARY LOAN COUNTER	UTM BORROW
Restricted to verified UTM members	—	—	●	●
Reputation earned from completed loans	—	○	—	●
Cryptographic proof of physical handover	—	—	○	●
Cryptographic proof of return	—	—	○	●
Tracked deadline with overdue alerts	—	—	●	●
Structured search by college and faculty	—	○	—	●
Moderation over an immutable audit trail	—	○	○	●
Student-owned items, no fee and no deposit	●	—	—	●

Assessed against the three channels a UTM student actually has available today.

● fully supported ○ partial or informal — absent

Marketplace apps sell; loan counters lend only what the university already owns.

Not one existing channel can prove that a physical handover actually happened. **UTM Borrow can** — with a signed, time-limited token that cannot be forged, replayed, or quietly undone.

Scan to open the live app
utmborrow.alsakkaf.site